The Norns III: Seeding Simulated-Annealing Placement with Provenance and Recovered Clusters How hot can the anneal start before it destroys a good initial placement?

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Abstract—Simulated annealing is the workhorse of FPGA placement, and it starts from an initial placement that the anneal then destroys and rebuilds. What if the initial placement carried real information – the HDL hierarchy a provenance-tracked front end kept, or the clusters a graph analysis recovered from the flat netlist? This paper asks whether such seeds improve placement, and, because the anneal's early phase can erase any seed, at what initial temperature the seed's advantage dies. The seeds come from Odin III's provenance hierarchy and from Loki III's L2 clusterings (Louvain, spectral, affinity propagation, k-means); the die-level arrangement of clusters comes from a cut-weighted quotient-graph layout; and the seeds enter VPR through its initial-placement and floorplanning-constraint interfaces, with VPR's stock checkpointing flag (on by default, following RLPlace) enabled throughout and an off sweep as the control arm. We report seed retention at scale 1 and at twice each design's crossover, quality of results against an unseeded baseline, and the largest tested temperature scale at which each seed still helps – the recipe a clustering-based initial placement should be used with. A second round adds bitwise seeds (Odin III word membership laid out as bit-sliced runs), a random start and a coarse-random arm, and two structure analyses that need no placement run; each hypothesis is reported with its deciding number, and an appendix keeps the research log. Rounds one and two are over 15 designs with one VPR seed per configuration. A third round asks whether VPR's packing step dilutes the HDL structure, builds a packer (a copy of VPR's AAPack clusterer with a provenance-affinity gain term) and tests it against the stock packer on 15 designs with three VPR seeds per configuration, adds the top-level I/O buses as words, and reads the annealed placement against provenance; its decision rules were committed before any measurement (one later amendment, to the proximity analysis, is recorded). Affinity packing raises word purity without growing the CLB count, but routed wirelength is slightly worse than the stock packer's and that comparison is inconclusive.

Index Terms—FPGA placement, simulated annealing, provenance, clustering, VTR

I. Introduction

VPR's simulated-annealing placer starts from an initial placement and anneals it: the schedule begins hot enough to move anything, and the early phase rearranges the circuit wholesale. The question this paper asks is whether the initial placement can be made to mean something, and how much of that meaning survives the anneal. Gharibian et al. showed that system-level structure recovered from a flat netlist correlates with placement locality [1]–[3]. Odin III keeps the provenance link that makes recovery unnecessary for the hierarchy [1], and Loki III recovers clusters from the flat netlist without a language model. RLPlace [4] showed that the anneal itself can be improved – directed moves chosen by a reinforcement-learning agent and checkpointing that prevents oscillation – starting from VTR's analytic placement. This paper connects those lines: the seed carries the structure, the anneal is measured for how much of it the seed can keep, and the largest tested temperature scale at which the advantage survives is reported as data, not tuned away.

The contributions are: (1) a seeding methodology – provenance hierarchy and four recovered clusterings, arranged on the die by a quotient-graph layout, entering VPR through its initial-placement and constraint interfaces, with VPR's stock checkpointing enabled and its off sweep as the control arm; (2) the crossover sweep, which calibrates the highest useful initial temperature per design against the best placement any seed achieves; (3) the retention and quality numbers per seed, with the full-seed versus inter-cluster-only split, which measures sensitivity to the intra-cluster order; (4) a second round that adds bitwise (word and bit-slice) seeds, a random-start and a coarse-random arm that separate the clustering from the coarse layout, and structure analyses of when seeding helps (Sections III-G and IV-A), with the research log in the appendix.

II. Background

A. Simulated annealing placement and RLPlace

VPR places by simulated annealing from an analytic initial placement [5], [6]. The anneal's moves are mostly random; its early phase can undo any structure the start carried. RLPlace improves the anneal rather than the start: a K-armed bandit selects among directed move

types (simple moves and swaps, bounded centroid, median, critical-path moves), and checkpointing periodically saves the best placement so oscillation cannot throw it away [4]. Our experiment keeps the anneal as-is (VPR already ships RLPlace-style checkpointing, on by default) and changes the start instead; the checkpointing prediction – that save-best/restore extends the useful temperature range – is one of the hypotheses we test.

B. The seeds

The provenance hierarchy (Urd) is the module structure Odin III records on every netlist object; it needs no recovery. Urd has a single cluster only on sha and stereovision3, two on diffeq1 and diffeq2 (their one module plus a flat group), and one per populated module plus a flat group (3 to 23) on the other designs, so its signal is degenerate on the two one-cluster designs. The recovered clusters (Verdandi) are Loki III L2 partitions: Louvain, spectral, affinity propagation and k-means at several k, scored in the Loki III paper [7] against the known hierarchy. Agent default: Norns builds the Loki L0 graph.json from the same Odin III BLIF, because Loki's key.json uses Odin API names that do not join to VPR atom names; the clustering code is Loki's. The two seed families differ exactly in the question's terms: provenance is the structure the designer wrote, clusters are the structure the flat netlist still shows.

III. Methodology

A. Seed construction

Each seed is a partition of the placed blocks. Its die-level arrangement comes from the quotient graph: contract each cluster to a node, weight edges by inter-cluster net cuts, lay the quotient out by a coarse force-directed solve, and scale it to the die. Inside a cluster, blocks take the seed's internal order, or a shuffle in the inter-cluster-only arm; the two arms differ only in that intra-cluster order, not in the coarse cluster layout.

B. Seed construction in detail

One die, one netlist. Every seed targets the same die: VPR's k6_frac_N10_frac_chain_mem32K_40nm architecture with six-input LUTs (K = 6). The netlist is the VTR-dialect BLIF that Odin III's front end writes for each design: read_verilog –nolatches –vtr-compat, proc, fsm, opt, flatten, opt -full, wreduce, hardmap, lower, abc –lut 6, stats, and write_blif –vtr –lut-k 6 –name-style provenance. Each configuration is one deterministic VPR run (–seed 1). The unseeded baseline run is the comparison point and also supplies the seed builders: its analytic initial placement (initial.place) gives every soft seed legal slots and its fine order, and its final.flat_place is the reference the retention metric uses.

Coarse layout. Every soft seed shares one die-level arrangement, the quotient-graph layout. Each cluster is contracted to one node and each pair of clusters is weighted by the number of nets cut between them; the weighted quotient graph is laid out by a force-directed solve (networkx.spring_layout, fixed seed), and the embedding is scaled onto the die's logic-slot bounding box. Each cluster then takes the free CLB slots nearest its laid-out point, nearest first. Clusters that exchange many nets therefore start close together, and the die fills in cluster-sized groups instead of a scatter.

Fine order. The soft arm has two variants that differ only inside a cluster. Full keeps the baseline analytic order: a cluster's blocks keep their relative analytic positions when they are assigned the cluster's slots. Inter shuffles the intra-cluster order under a fixed shuffle seed, so only the inter-cluster structure survives; because the coarse cluster layout is identical in the two variants, the pair isolates the fine order. The hard arm has no fine-order choice: it hands VPR one PartitionRegion per cluster, sized from the full seed's slots and expanded until it holds the cluster's logic (and nudged to cover a memory or multiplier site the cluster needs), and VPR's placement orders the blocks inside each region.

Placement macros. Every seed honours the die's placement macros: I/O pads never move (their I/O sites are fixed by the architecture), mult_36 macros stay anchored on their tile footprint, and an adder carry chain moves as one vertical run in a single column, head at the top (VPR places a chain as a direct vertical macro). Chains are assigned slots before the nearest-first fill, so a chain is never split.

The seed families. The Urd seed is the provenance hierarchy made explicit: Odin III writes each block's HDL module into its netlist name, the seed builder reads that module from the BLIF, and each populated module becomes one cluster; blocks that carry no module name join one flat cluster. Urd is one cluster on sha and stereovision3, two on diffeq1 and diffeq2 (one module plus the flat group), and one per populated module plus the flat group on the other designs. The rationale is the designer's locality: modules are the units the designer grouped, and logic inside a module was written to communicate (shared words, state, control), so a module cluster concentrates the connectivity the placer is paid to shorten. That is the hypothesis the Urd arm tests.

The Verdandi seeds are Loki III's L2 clusterings of the flat netlist: Louvain, spectral, affinity propagation, and k-means at k, $2k$ and $4k$, with k the design's module count (two where the design has one module). Norns builds the L0 graph.json the clusterings read from the same Odin III BLIF, because Loki's key.json uses Odin API names that do not join to VPR atom names; the clustering code is Loki's. A recovered cluster approximates the provenance grouping from connectivity alone, so comparing Urd with Verdandi isolates what the HDL structure adds over the graph.

Why a seed should beat a random start. A random placement scatters communicating logic across the die; the quotient layout puts each cluster's logic together and

puts clusters that exchange nets next to each other, so the anneal starts from the structure it is meant to find and refines from there. The experiment tests that mechanism rather than assuming it, and it compares against VPR's analytic start, which already respects connectivity, not against a random placement. Two measurements report what the seed did: retention, the fraction of a seed's CLBs found inside its seed cluster's region at exit, has medians 0.68 (full) and 0.70 (inter) at VPR's own start temperature and 0.52–0.74 per family at twice the crossover; and the crossover sweep finds the largest tested temperature scale at which the seed still wins, the point where its contribution stops surviving the anneal.

What the HDL constructs say about a good start. HDL modules, buses, datapath words and finite-state machines group logic that communicates: a bus is one word fanned out to every consumer, a datapath chain is a dataflow line, and an FSM's next-state logic and registers exchange signals every cycle. Provenance preserves that grouping in the netlist, so a seed built from it starts the placer with the designer's locality already on the die; L2 clusters try to rediscover it from connectivity alone, and this experiment measures how well that substitute places, not how closely the clusters match the modules. That is why Urd and Verdandi are the arms, and why the paper asks whether recovered structure is interchangeable with provenance.

C. Injection

Seeds enter VPR two ways, both already in the pinned build (v8.0.0+18416): soft, as an initial placement file (–read_initial_place_file) that the anneal is free to move away from; and hard, as floorplanning constraints (–read_vpr_constraints) with one PartitionRegion per cluster, which the anneal cannot violate at any temperature. The hard arm constrains the regions only, so the anneal chooses the fine placement inside each one (one hard variant, no full/inter split). –write_initial_place_file dumps VPR's own analytic start for the baseline study. VPR's placement macros are respected: I/O pads and the multiplier tiles stay anchored, and carry chains move as one vertical run. The seeds are built from an Odin III provenance BLIF and Loki III's clusterings of the same netlist, arranged by a cut-weighted quotient-graph layout with networkx (spring layout, fixed seed). Building the seed files also needs VPR's unseeded baseline placement for legal slots, so each design's baseline run precedes its seeds.

D. The crossover sweep

The calibration loop follows the data, in order. First, run every seed and keep the best final placement P^* (and which seed made it). Second, cold-start a fresh anneal from P^* and sweep the initial temperature upward; the crossover is the largest tested scale at which the cold start still routes shorter than the unseeded run at the same scale, a lower bound on the true fall point when the sweep stops at its cap. The sweep is run once per design, so S is design-level and identical across that design's seeds. The temperature is reported as a multiple S of VPR's automatic start-temperature estimate. Third, run every seed at twice the design's crossover against an unseeded run at the same temperature: whether the seed still wins there tests how much of its advantage survives a hot restart. The checkpointing arm repeats the sweeps with VPR's checkpointing disabled; per RLPlace, the crossover should move hotter when it is on.

E. Metrics and data

Per run: seed retention (fraction of cluster CLBs still inside their seed cluster's region at exit; mean displacement from the seed coordinate), routed wirelength, critical-path delay, minimum channel width, placement runtime, and the per-temperature cost curve. Designs are the 15 VTR benchmarks that fit the development machine, smallest first, one run at a time under the heavy lock. Every number in this paper traces to a kept record; large data keeps its scripts.

F. Benchmarks

Table I reports each design's construction and its post-technology-mapping size. The module count is the front end's, while the CLB count and the grid are VPR's own numbers for the packed netlist, so the table describes the designs as the experiment actually places them.

The 15 designs span two orders of magnitude after technology mapping: from 13 CLBs (ch_intrinsics) to 1956 (stereovision2), 4843 CLBs in total. The mean is 322.9 CLBs and the median 125; the mean sits well above the median because three stereovision designs carry most of the logic. VPR auto-sizes a square grid to the packed netlist, with a mean side of 27.2 tiles (median 20) and a mean of 1076.4 grid tiles (median 400). Grid size follows the largest resource demand, not the CLB count: I/O-heavy and memory-heavy designs such as mkDelayWorker32B get a large die for few CLBs.

G. Round two: bitwise seeds and structure analyses

Round one seeded VPR with module-level clusters. Datapath logic is bit-level: a 32-bit register or adder is a regular stack of identical bit cells, and the carry chain already shows that VPR places such stacks as ordered vertical runs. Round two therefore adds a bit-level seeding aspect that keeps the module hierarchy at the coarse level (Section A records where the idea came from).

a) Word membership from Odin III data, not names.: The obvious source, the provenance BLIF's block name module/$nN@file:line, is too coarse: the line is the HDL statement, so one always block is one group (the largest (module, file:line) group in any design holds up to 5248 blocks, and sha's holds 1739 of its 2506 blocks; 370 of the 790 statement lines that hold word blocks hold more than one word, up to 320). The finer records exist in the

TABLE I
Benchmark construction and post-technology-mapping size. All fifteen designs are from the VTR 8.0 benchmark set (vtr_flow/benchmarks/verilog). Modules is the front end's module count, CLBs the clb blocks VPR packs, Memory and Mult the memory and mult_36 blocks, and Grid the auto-sized device grid with its tile count in parentheses. Sizes are read from each design's baseline run log and kept in data/benchmark-sizes.json.

Design	Suite	Modules	CLBs	Memory	Mult	Grid (tiles)
ch_intrinsics	VTR 8.0	2	13	1	0	10×10 (100)
stereovision3	VTR 8.0	1	15	0	0	7×7 (49)
diffeq2	VTR 8.0	1	22	0	5	16×16 (256)
diffeq1	VTR 8.0	1	37	0	5	16×16 (256)
mkPktMerge	VTR 8.0	7	21	15	0	28×28 (784)
raygentop	VTR 8.0	15	143	1	6	19×19 (361)
sha	VTR 8.0	1	142	0	0	17×17 (289)
boundtop	VTR 8.0	13	54	1	0	15×15 (225)
mkSMAdapter4B	VTR 8.0	8	125	5	0	20×20 (400)
or1200	VTR 8.0	16	252	2	1	27×27 (729)
stereovision0	VTR 8.0	25	707	0	0	33×33 (1089)
stereovision1	VTR 8.0	16	788	0	43	40×40 (1600)
mkDelayWorker32B	VTR 8.0	16	74	43	0	50×50 (2500)
stereovision2	VTR 8.0	24	1956	0	192	82×82 (6724)
blob_merge	VTR 8.0	2	494	0	0	28×28 (784)

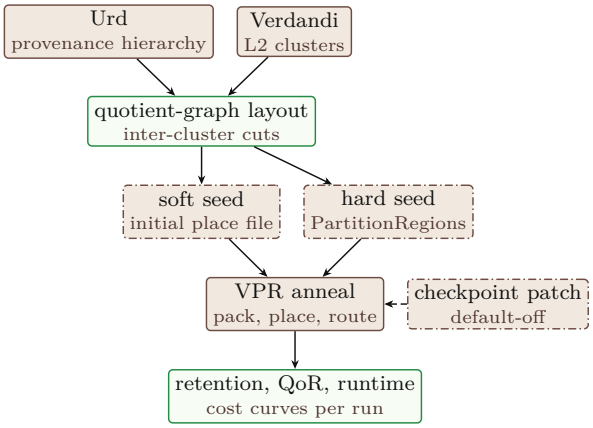


Fig. 1. The Norns III experiment. Seeds from provenance and recovered clusters, arranged by a quotient-graph layout, enter VPR softly (initial placement) or hard (constraints). The crossover sweep cold-starts from the best placement and raises the initial temperature until the seed's advantage dies or the sweep reaches its cap; the checkpointing arm (VPR's stock flag, on vs off) tests whether save-best/restore extends the range.

Odin III session that writes the BLIF, so the seed builder reruns the identical script, checks that the BLIF it writes equals the kept one byte for byte (all 15 designs match), and reads the records Loki III's labeller reads: the HDL name bus[i] of a cell's output net and the source spans of each cell, matched to the word-level cells that exist before technology mapping. A final cell joins its BLIF block through the net id the BLIF name carries ($n<id>), so the join is by identifier, not by string (1204 of the 106262 Odin III final cells, 1.1%, found no BLIF block and are left unlabelled, from none to 22.2% per design, the worst being mkDelayWorker32B with 942 of 4235; the writer legalises dotted names, handled explicitly). A block's word is the first rule that applies: (bus) an output net named bus[i] puts it in word bus at bit i, which holds for every flip-flop that keeps a register name; (cell) exactly one word-level cell's spans lie inside the block's spans, which is true for adders, multipliers, memories and single-statement LUTs; otherwise (none) the block has no word. Bit order inside an adder word is the carry-chain position. ABC's cone provenance merges the spans of a LUT that mixes words, and its edge budget makes some runs share one record, so 49% of the LUTs get a word against 100% of the flip-flops and 99% of the hard cells. Over all designs, 80.6% of the blocks are in a word (per-design median 66.7%); 5190 words in all.

b) Seed families.: A CLB joins a word cluster when at least one third of its atoms carry a word and that word is the plurality; otherwise it keeps its Urd module cluster. Four families follow, each entering VPR as a soft initial placement at VPR's own start temperature (scale 1). words: the word CLB groups are the clusters, arranged by the round-one quotient-graph layout. slices: the same clusters, but each word is laid out as a vertical run in bit order (bit 0 on top, the carry-chain direction), runs of equal length aligned to the same rows, so that bit i of connected words shares a row; true carry chains are untouched. hier: Urd module clusters at the coarse level and, inside each module, the words as bit-sliced runs. words_partial: only the word CLBs are seeded, every other CLB keeps VPR's analytic position (and every carry chain with an unseeded CLB stays whole). The words family also has the inter-cluster-only variant. Pads and multipliers stay anchored as in round one.

c) Random and coarse-random arms (RQ-N4).: random is a random legal start: every logic CLB takes a random slot and carry chains a random window, fed as a soft seed at scale 1. The coarse-random arm keeps the clusters and their within-cluster order but permutes the

cluster positions on the die (urd_shuf for Urd, and the *_shuf twin of the best new family).

d) Structure analyses (no VPR runs).: RQ-N1 measures, on each design's unseeded baseline, how module- and word-coherent three placements are: VPR's analytic start, its final placement, and a random null (logic-CLB coordinates permuted, 20 samples). A logic CLB carries the plurality label of its atoms (module: the Urd module; word: the word with most atoms in the CLB; a CLB without one is unlabelled). Coherence is the NMI between a $g \times g$ grid partition of the labelled CLBs ($g = \max(2, \lceil\sqrt{\#\text{labels}}\rceil)$, the grid spanning the labelled CLBs' bounding box) and the labels. The compactness is a separate metric that does not use the grid: the CLB-weighted mean over labels with at least two CLBs of the label's CLBs divided by the area, in tiles, of the label's bounding box. A labelling is scored only with at least four labelled CLBs and two labels, and a design is counted against the null only when the null's standard deviation is positive (counted separately for the two metrics). A design with fewer than two real Urd modules (diffeq1 and diffeq2, which have one module plus the artificial flat group, and sha and stereovision3, which have none) has no module row. Label sets with fewer than 20 labelled CLBs are flagged in the results. RQ-N2 correlates the seed gain with three structure measures of the netlist graph: Louvain modularity Q; the z-score of Q against a degree-preserving null (double edge swaps, 10 per edge, 20 samples); and the inverse Rent exponent $-p$, with p fitted from the terminals of BFS balls of geometric sizes (a lower exponent is a more local design). Spearman's ρ and p-value are reported with $n = 15$, which is small, and again as partial correlations with the edge count regressed out, because graph size drives both $z(Q)$ and the gain.

e) Hypotheses and decision rules.: Paired comparisons are per design (variant over reference), reported as a geomean ratio, the number of designs where the variant is better and a two-sided exact sign test. H7a: words beats Urd in soft wirelength (supported if the geomean is below 1 and $p < 0.05$, not supported if the geomean is at least 1, else inconclusive). H7b: slices beats words on wirelength and channel width (supported if both geomeans are below 1 and the wirelength sign test has $p < 0.05$). H7c: hier has the lowest geomean wirelength of all eleven soft families. H7d: words_partial is better than the full word seed (supported if the geomean ratio is below 1 and the wirelength sign test has $p < 0.05$, not supported if it is above 1 with $p < 0.05$, else inconclusive; no equivalence margin was fixed, so "at least as good" is not testable here). H1: the baseline's final placement is module- and word-coherent (supported if, for both labellings, at least two thirds of the designs lie more than two null standard deviations above the null in NMI and at least two thirds in compactness, the two counted separately; not supported if fewer than a third do for both labellings and both metrics). H2: structure predicts gain (supported if at least two of the three measures correlate positively with the best round-two gain at $p < 0.05$; not supported if no correlation of either sign reaches $p < 0.05$). H4: random start > permuted clusters > quotient layout ≥ analytic start in wirelength ratio, for Urd and for the best new family.

f) Which rules predate the data.: These rules were not registered anywhere outside the repository, so their order is stated from the commit times of that repository. The decision-rule code for H7a to H7d and H4 was committed at 15:21 on 2026-10-24 and H1's rule (with the compactness criterion) at 15:24, before the first round-two VPR data (15:38). Some of those rules were later found weak and changed; the appendix lists the changes. After the first data: tie handling in the sign tests (ties left out) came with the first data commit; the methodology text that states the rules was written at 17:31, after nine designs; the best family was frozen at 13 designs after their results were seen (so the hard, crossover and permuted arms of the best family use a choice made on data); the H2 rule was briefly changed and reverted, and the H7d rule was replaced after review. The next round fixes every decision rule, and the equivalence margin of H7d, in a commit before any run.

H. Round three: packing, I/O words and placement proximity

Round two found every soft seed within a few percent of the baseline. Round three asks whether the packing step, which runs before placement and decides which atoms share a CLB, is where the HDL structure is lost, builds a packer that keeps it, and then asks what the annealed placement itself says about provenance. The two questions that started it are quoted in the research log (Section ??).

a) The Norns packer.: VPR's packer is AAPack, the greedy clusterer in vpr/src/pack (not T-VPack); in VTR's default analytical flow the same clusterer runs as APPack, with candidate gains attenuated by distance in the global placement. The Norns packer is a copy of the changed files of that clusterer (VTR commit 3c9a4d23b2, MIT licence kept) with one added term: an atom carries a group id read from a file (–norns_affinity); a cluster's dominant group is the plurality group of its atoms so far; a candidate molecule gets $+\beta$ (–norns_affinity_weight) for every atom in that group, and up to 30 unclustered members of the group are proposed as candidates, because the bits of a word are parallel and no net links them. The stock per-atom gain lies in $[0, 1]$, so β is in the units of one connected atom's gain. Legality stays with AAPack's legalizer. The change is 7 files, 308 added lines and 0 removed. Control gate: with the option absent and with $\beta = 0$, the packer must write a .net byte-identical to the stock vpr's in –pack and in the analytical flow on all 15 designs, before any experiment run: 60 comparisons over 15 designs, all byte-identical; the same $\beta = 0$ check for the module and word+io affinity files (–pack, seed 1): 30 of 30 byte-identical.

b) Limits of the proposal step.: The step hands up to 30 unclustered members of a cluster's dominant group to the candidate queue, scanning the group in netlist (atom id) order. Its counter counts the molecules handed over, not the molecules the queue accepted, so the cap of 30 can be spent on molecules the queue ignores (already queued, or beyond APPack's distance threshold), and the scan always starts at the lowest atom ids of the group: which members are tried first is biased by netlist order. The behaviour is unchanged in this paper (Stage-4 results stand); it was measured with a log-only counter added to the packer, which leaves the packing unchanged (45 of 45 .net files byte-identical to the previous binary's, 45 of 45 flat placements identical to the Stage-4 run; pack-only style runs of the headline configurations, seed 1, no routing). Share of handed-over molecules that entered the queue, median over designs (module / word / word+io): 48.8% / 46.4% / 47.6% (pooled over all calls: 47.4% / 53.1% / 52.6%); proposal calls (one per cluster) that queued nothing: 13.1% / 54.6% / 49.2% (Table XIII). The fix (count only queued molecules, return a bool from the queue step, a scan order not tied to atom ids) is queued for round 4.

c) RQ-N8, dilution (no VPR).: On the stock packings of round one, per design: purity (plurality share of the labelled atoms of a logic CLB, atom-weighted), and the split factor (CLBs a word spans over the fewest CLBs that could hold it, 1.0 being best), at word and module level, against a null that assigns the same atoms at random to CLBs of the same sizes (50 samples). Round two's word coverage loss is split into the front end (atoms with no word) and packing (word atoms not in a CLB where their word is the plurality word under the 1/3 rule).

d) RQ-N10, I/O words.: A top-level bus name[i] is an io-word; its identity is propagated inward over LUT and flip-flop atoms, two hops past the atom next to the pad, through nets of fanout at most 16, giving an atom that has no round-two word the io-word it reaches from exactly one bit of exactly one io-word.

e) RQ-N9, packing experiments.: Arms: the stock packer, and the Norns packer with affinity source module, word or word+io (word, else io-word) at three values of β each, with the analytic start or with the round-two words_partial seed rebuilt on that run's own packing, three VPR seeds (1 to 3) per configuration. The grid was chosen on a pilot of the four smallest designs that reads only purity and CLB count (never wirelength): β_m is the largest value in $\{0.05, 0.1, 0.2, 0.5, 1, 2, 4\}$ whose median CLB-count growth is at most 5%, and the grid is its neighbours (module: $\beta_m = 4$, grid 1, 2, 4; word: $\beta_m = 4$, grid 1, 2, 4; word+io: $\beta_m = 4$, grid 1, 2, 4). The pilot did not discriminate between the betas. The median logic-CLB growth over the four pilot designs was negative at every β of the list (largest value -3.8%), so the 5% growth rule passed all of them and took the end of the list, and the purity gain of the word sources was the same at $\beta = 2$ and $\beta = 4$ (module: zero throughout). The grid $\{1, 2, 4\}$ is therefore not informed by the pilot, and a larger gain scale is untested. Each claim is reported on all 15 designs and without the four pilot designs; a status that differs between the two is called inconclusive (pilot-sensitive).

f) RQ-N11, proximity.: On the stock-packing, analytic-start runs, logic CLB pairs by relation (same word, same module, same statement, net-connected, same io-word) against the direct net count, which is stratified out: the stratified mean distance difference of related over unrelated pairs, as a fraction of the unrelated distance (negative is closer). A stratum counts only with at least 5 related and 30 unrelated pairs. The status is computed under the protocol text (4fb8b60: no design excluded, the sign rule needs at least 12 of the 15 designs, so a relation defined for fewer designs cannot be supported), which is the primary status, and the following variants are reported beside it, labelled. (i) The sign rule scaled to $\lceil 12n/15 \rceil$ designs, which the first analysis code used; the protocol does not register it. (ii) The amendment d805789: a design is left out of a relation when 95% or more of the sampled pairs are related (applied to the module relation and to the word relation, per design on the mean over the three seeds, with the scaled and the literal count), and the label permutations are 200 per design and seed. The primary permutation test is the protocol's 1000 label permutations pooled over designs; it is reported, not used in a status. The entry of 2026-10-25 in docs/plan/round3.md records these deviations.

g) Decision rules and their provenance.: Every hypothesis, metric, arm and numeric rule (including the margins $\pm 2\%$ and 1% for wirelength, 5 percentage points of purity, 5% CLB growth) is in docs/plan/round3.md, committed as 4fb8b60 on 2026-10-24 21:26 (UTC-4), before any round-three measurement or VPR run. Two later dated entries exist and the earlier entries are not edited. The entry d805789 (2026-10-24) is titled a clarification, but it is an amendment: besides definitions it changed two H11 rules (the label permutations, 1000 pooled over designs, became 200 per design and seed; a new exclusion of designs with 95% or more related pairs). It was committed at 22:27:05, before any Stage-4 result was analysed, but the Stage-4 runs of ch_intrinsics (60 runs, finished 22:26:38) and 40 of the 60 of stereovision3 had already finished. The entry of 2026-10-25 (the fix round after review) records what the analysis code did beyond the protocol text. Where the code deviated, the primary status in this paper is the one under the text of 4fb8b60 and the deviating variant is reported alongside, labelled.

IV. Results

The 15 completed designs give the ratios of Table II. Soft seeds stay within a few percent of the unseeded baseline in routed wirelength (1.009 for the full seed, 1.015 for the inter-cluster-only arm, both pooled over all seven

TABLE II
Geomean quality ratios vs the unseeded baseline, per seed family, for the soft initial-placement arm; the inter-cluster-only row shuffles the order inside each cluster and pools all seven seed families, and the k-means row pools k, $2k$ and $4k$. Below 1.0 is better. The hard-constraint arm is discussed in the text.

Seed	Wirelength	Delay	Chan. width	Runtime
Urd (hierarchy)	1.029	1.004	0.958	1.068
Louvain	1.008	0.982	0.980	1.091
Spectral	1.006	0.993	0.968	1.098
Affinity propagation	1.002	1.003	0.982	1.086
k-means (k, $2k$, $4k$)	1.006	0.985	0.975	1.088
Inter-cluster only	1.015	0.981	0.987	1.102

TABLE III
The crossover sweep, per seed family: the largest tested scale S of VPR's automatic start-temperature estimate at which the cold start from P^* (the best placement over all seeds) still routes shorter than an unseeded run at the same S, and the seed's median retention at the hot step ($2S$). S is design-level (one sweep per design), so it is identical across families; where the sweep stopped at its cap, S is a lower bound.

Seed	Crossover S	Retention
Urd (hierarchy)	16×	0.744
Louvain	16×	0.520
Spectral	16×	0.635
Affinity propagation	16×	0.564
k-means (k, $2k$, $4k$)	16×	0.667

seed families) while cutting the minimum channel width to 0.973 and costing 1.087 in placement runtime (both also pooled). The hard arm is worse on average (geomean 1.088 over the 104 runs that completed; 14 of those still beat the baseline, and stereovision3's hard geomean is 0.906), but not uniformly: one blob_merge k-means$_{2k}$ run is missing from the arm because VPR's initial placement could not fit 9 CLBs inside its regions (the constraints are too tight, not a packing failure). Fixed regions may block the anneal's fine moves across cluster borders – a hypothesis the aggregate numbers are consistent with. No single seed family dominates: Urd is the worst full seed in soft wirelength (1.029) and the best hard seed (1.062 against 1.075-1.120 for the recovered families). Retention at scale 1 is 0.68 of a seed's cluster CLBs (full) and 0.70 (inter).

A. Round two results

RQ-N7, bitwise seeds (15 designs, one VPR seed each). H7a is inconclusive: words over Urd is 0.986 in wirelength, better on 10 of 15 designs (sign test $p = 0.302$). H7b is inconclusive: slices over words is 0.982 in wirelength (better on 8 of 15, $p = 0.791$) and 1.033 in channel width (better on 6 of 15). H7c is not supported: hier ranks 8 of 11 families at 1.014; the lowest is words_partial at 0.994. H7d is inconclusive (no evidence of a difference): words_partial over words is 0.980 (better on 9 of 15, $p = 0.607$). Ties (identical results) are left out of the sign tests: 0 for H7a, 1 for H7b and 0 for H7d. A family

TABLE IV
Round two: geomean ratios vs the unseeded baseline at scale 1 for the soft arms (Urd from round one for reference). The last column is the number of designs where the routed wirelength is below the baseline. WL is routed wirelength, Chan. the minimum channel width, Time the placement runtime, T_0 the anneal's start temperature that VPR derives from the starting placement. Below 1.0 is better (except T_0, a diagnostic).

Seed	WL	Delay	Chan.	Time	T_0	Better
Urd (round 1)	1.029	1.004	0.958	1.068	3.23	3/15
Words	1.015	0.980	0.964	1.254	6.24	5/15
Words, inter only	1.017	0.984	0.970	1.264	7.35	6/15
Slices	0.996	0.986	0.995	1.268	8.07	9/15
Hier	1.014	1.002	0.982	1.247	4.52	6/15
Words, partial	0.994	0.980	0.962	1.257	4.63	7/15
Random start	0.997	0.978	1.012	1.275	9.24	8/15
Urd, permuted	1.020	0.992	0.972	1.227	3.09	4/15
Words, partial, permuted	0.980	0.972	0.988	1.179	5.19	7/15

with nothing to rearrange reproduces its reference exactly (a design whose words are all single-CLB, or that has one module): slices equals words on 1 design(s), hier equals Urd on 1 and words_partial equals words on 0. Seed audit. Rebuilding the seven round-two seed families from the kept word membership with the committed builders gives seed files byte-identical to the ones VPR ran from on all 15 designs; so every slices run used the column-major bit-order fallback of the committed code, and that fallback changes the slices seed on 6 designs (ch_intrinsics, diffeq1, mkPktMerge, raygentop, stereovision0, stereovision2). The designs where slices equals words as a seed file (mkDelayWorker32B) and where hier equals Urd (mkDelayWorker32B) are equal because there is nothing to rearrange, not because a code path was skipped. On mkDelayWorker32B only 9 CLBs carry a word label, in 7 words, and no word has two CLBs off the carry chains.

words_partial places only the word CLBs, so how much of each design it moves matters for H7d (Table XIV). Of the logic and memory CLBs the seed moves a median of 79% per design (6 of 15 designs at 90% or more, down to 15% on mkDelayWorker32B); counting every placed block, pads and multipliers included, which are anchored in every arm, the median is only 18% (for example 11 of 243 on ch_intrinsics). Where the seed covers nearly all logic CLBs, words_partial differs little from words, so H7d compares two similar placements; where it covers little (for example mkDelayWorker32B), most of the seed is VPR's own analytic start. H7d is therefore a weak test of "seeding fewer CLBs" on the designs where the share is high. The best new family by geomean wirelength is words_partial (chosen on the 13 designs whose soft arms were complete when the hard and crossover arms were queued), so the coarse-random, hard and crossover arms below use it. Its geomean of 0.994 is a selected figure: the family was the best of four on those 13 designs and is reported on them (and on the 2 added later), so it is

biased low, and it lies inside the band of the other arms. Its hard arm (one region per cluster, scale 1) is 1.141 in wirelength over 12 designs, better than the baseline on 2; round one's hard arm pooled to 1.088 (all families, all designs). That figure is over the 12 survivors only, and the 3 designs that failed are among the largest, so survivors flatter it. Round one's Urd hard arm over the same 12 designs is 1.042 (better on 3): on equal designs the bitwise family's hard arm is worse than Urd's. The hard arm failed in VPR's initial placement on 3 designs (stereovision0, stereovision1, stereovision2: no free sites for the constrained blocks, the same failure round one had on blob_merge) and those are left out. The builder's region check counts a box as large enough when it holds as many logic sites as the cluster's own CLBs; it does not count the unseeded CLBs that this family leaves at their analytic positions, which can sit inside a box, so these failures may be builder-side rather than a property of hard regions. Its crossover sweep, started from its own scale-1 placement, has median crossover scale 16 against 16 for round one's best-of-seven P^* over the 13 of 15 designs where it won at some scale (at least round one's on 11 of 13); at twice the crossover the retention median is 0.20 and the seed beats an unseeded run at the same temperature on 5 of 13 designs. stereovision3 and mkPktMerge had no winning scale at all and are left out of those medians and counts (round one won up to 256× on stereovision3 and 16× on mkPktMerge). Counting them as failures, the family is at least round one's on 11 of 15 designs, and the median over all 15 (a failure taken as below the lowest scale) is 16 against 16.

RQ-N4, clustering versus coarse layout. H4 is not supported. For Urd the wirelength ratios are random start 0.997, permuted cluster positions 1.020, quotient layout 1.029; for words_partial they are 0.997, 0.980 and 0.994. The analytic baseline is 1 by definition, so a random start below 1 means the anneal from a random start ended with shorter wire than from VPR's analytic start; the gap between the permuted and the quotient arm is what the coarse layout is worth beyond the clustering itself. Permuting the cluster positions changes the wirelength by -0.9% for Urd and -1.4% for words_partial relative to the quotient layout (negative is shorter).

RQ-N1, does VPR rediscover the hierarchy? H1 is inconclusive. Table V gives medians over designs: the final placement is above the random null in NMI on 8 of 11 designs for modules and 9 of 15 for words, and in compactness on 10 of 11 and 14 of 15 (more than two null standard deviations). Modules are counted over 11 designs, not 15 or 13: diffeq1 and diffeq2 have a single module, whose second label is the artificial flat group, and sha and stereovision3 have none. Label sets with fewer than 20 labelled CLBs carry little information (module labels on ch_intrinsics; word labels on ch_intrinsics, stereovision3, mkDelayWorker32B). Final and analytic placements are close, and the final NMI is at least the analytic one on 5 of 11 (modules) and 7 of 15 (words) designs. With many small labels the NMI of a random placement is itself high (word null 0.66), so compactness is the cleaner signal.

TABLE V
RQ-N1: medians over designs of the grid-partition NMI and label compactness of the analytic start, VPR's final placement and a random null (20 permutations). Designs with fewer than two real modules have no module label and are left out of that row.

Labels	n	NMI (start/final/null)	Comp. (start/final/null)
Module	11	0.26/0.31/0.12	0.20/0.26/0.09
Word	15	0.72/0.73/0.66	0.33/0.32/0.08

TABLE VI
RQ-N2: Spearman ρ (p) between per-design structure and seed gain (1 - wirelength ratio): Urd round one, best of seven round-one families, best new family. The inverse Rent exponent is $-p$.

Gain	Q	$z(Q)$	$-p_{\mathrm{Rent}}$
Urd, round one	-0.17 (0.533)	-0.44 (0.098)	0.12 (0.657)
Best round one	-0.09 (0.742)	-0.23 (0.413)	-0.16 (0.558)
Best new family	-0.36 (0.191)	-0.58 (0.023)	0.13 (0.648)

RQ-N2, does structure predict the gain? H2 is inconclusive: 0 of 9 correlations are positive at $p < 0.05$ and 1 negative (the hypothesis predicts positive; no multiple-comparison correction; $n = 15$ designs, so only a large ρ can be told from chance). Table VI lists Spearman's ρ (and p).

Graph size is a confounder. The measure $z(Q)$ rises with the number of edges ($\rho = 0.85$ ($p < 0.001$)), and the gains fall with it: the best new-family gain against edges is $\rho = -0.60$ ($p = 0.017$), the best round-one gain $\rho = -0.29$ ($p = 0.289$), Urd's $\rho = -0.37$ ($p = 0.177$). Table VII repeats the correlations of Table VI as partial Spearman correlations with the rank of the edge count regressed out of both variables. The null graphs behind $z(Q)$ use 10 double edge swaps per edge on every design (an earlier run capped the swaps at 200000, which left 3.8 to 9.8 per edge on the five graphs above 20000 edges; the cap was removed and the null recomputed). The best-of-k gains are maxima over k families per design ($k = 7$ round one, $k = 4$ round two), a positively biased gain, and the noise of a single VPR seed per configuration is unquantified.

B. Round three results

RQ-N8, does packing dilute structure? H8 is supported: over the 15 stock packings the median word purity (plurality share) is 0.63 against 0.20 for a random grouping of the same atoms (higher on 15 of 15), and the median split factor is 4.00 (above 1 on 12 of 15). The median module purity is 0.99, so module affinity has little to gain on most designs. Of the logic atoms, 65.8% carry a word (median); 34.2% lack one (front end: ABC's cone provenance, mixed spans 33.8% and no match 0.4%) and a further 20.7% carry a word but sit in a CLB where

TABLE VII
RQ-N2 with graph size controlled: partial Spearman ρ (p) between structure and seed gain, the rank of the edge count regressed out of both.

Gain	Q	$z(Q)$	$-p_{\text{Rent}}$
Urd, round one	0.11 (0.710)	-0.27 (0.358)	0.18 (0.543)
Best round one	0.15 (0.607)	0.04 (0.891)	-0.14 (0.634)
Best new family	0.09 (0.761)	-0.16 (0.574)	0.24 (0.401)

TABLE VIII
Round three, headline configurations (the β_m of each source): geomean ratio over the 15 completed designs against the stock packer with the same start, same VPR seeds. Below 1.0 is better; CLBs is the number of logic CLBs; Pack and Flow are the packing and total VPR runtimes. The stock arm ran on the stock vpr and the Norns arms on the Norns vpr, two builds, so Pack and Flow are not a same-binary comparison (the same-binary packing time ratio is in the text). The seed arm's stock reference is the stock packing with the same words_partial seed rebuilt on it.

Arm	WL	Chan.	Delay	CLBs	Pack	Flow
module, analytic	1.051	1.061	1.017	0.965	1.318	1.084
word, analytic	1.032	1.034	1.002	0.985	1.125	1.014
word+io, analytic	1.030	1.036	0.996	0.984	1.130	1.005
module, word seed	1.064	1.063	1.031	0.965	1.315	1.100
word, word seed	1.039	1.042	1.009	0.985	1.116	1.039
word+io, word seed	1.039	1.042	1.007	0.984	1.127	1.052

their word is not the plurality word or fails the 1/3 rule (packing).

RQ-N10, I/O words. H10a is supported: the median LUT word coverage rises from 31.8% to 53.9% with io-words (median gain 24.1%, positive on 15 of 15 designs).

Without the four pilot designs. H8, H10a and H11 use the stock packings and no beta, so the pilot, which only chose beta, cannot affect them; the n=11 pass is reported beside the n=15 status, which stays the primary one. H8 on the 11 designs: supported (median purity 0.59, split 4.00). H10a on the 11 designs: supported (median gain 22.3%, positive on 11 of 11; the rule's 12 of 15 scaled to 9, not registered).

Seed noise. The stock packer's routed wirelength varies across the three VPR seeds with a median coefficient of variation 2.4% (maximum 9.7%, n=15 designs), which is the floor under any wirelength difference below a few percent.

module affinity. H9a is inconclusive (pilot-sensitive): the module-level purity moves by 0.000 (median, n=15; 0.035 without the pilot designs) with a median change of -1.6% in logic CLBs (largest 13.3%). H9b is inconclusive (pilot-sensitive): with the analytic start the routed wirelength is 1.051 of the stock packer's (better on 3 of 15; 90% interval 1.020 to 1.082; 1.069 without the pilot designs, status there supported (higher WL)). H9c is not supported: with the word seed it is 1.064 (better on 2 of 15).

word affinity. H9a is supported: the word-level purity moves by 0.111 (median, n=15; 0.120 without the pilot designs) with a median change of 0.0% in logic CLBs (largest 13.3%). H9b is inconclusive (pilot-sensitive): with the analytic start the routed wirelength is 1.032 of the stock packer's (better on 4 of 15; 90% interval 1.006 to 1.062; 1.053 without the pilot designs, status there supported (higher WL)). H9c is not supported: with the word seed it is 1.039 (better on 4 of 15).

word+io affinity. H9a is supported: the word+io-level purity moves by 0.104 (median, n=15; 0.124 without the pilot designs) with a median change of 0.0% in logic CLBs (largest 13.3%). H9b is inconclusive (pilot-sensitive): with the analytic start the routed wirelength is 1.030 of the stock packer's (better on 4 of 15; 90% interval 1.005 to 1.057; 1.050 without the pilot designs, status there supported (higher WL)). H9c is not supported: with the word seed it is 1.039 (better on 5 of 15).

I/O words in the affinity (H10b). not supported: word+io over word is 0.998 (better on 4 of 15, $p = 1.000$).

Packing time on one binary. The stock arm of Stage 4 ran on the stock vpr and the Norns arms on the Norns vpr, two builds, so the ratios of Table VIII compare different binaries (and the Flow column includes routing).

Primary same-binary measurement (pack-only style, no routing, seed 1 to 3, 180 runs): on the Norns vpr, for each design and seed the stock arm (option absent) and the three Norns arms ran back to back, so any load affects the arms alike. The geomean over designs of the Norns to stock packing time ratio is 1.305 / 1.120 / 1.129 (module / word / word+io). These runs took place 2026-10-25 10:42:54 to 2026-10-25 11:05:44 (UTC-4) with a concurrent load: another lane ran CPU work outside the heavy lock (the one-minute load average at the start of a run had median 1.2, maximum 4.8). Packing times of the small designs are a few tenths of a second, so the ratio is coarse and approximate.

Not interleaved, hence approximate: the Stage-4 Norns packing times over a separate rerun of the stock arm on the Norns vpr (45 runs, 45 flat placements identical to the Stage-4 stock run, taken at another time and load) give 0.744 / 0.635 / 0.638, against 1.318 / 1.125 / 1.130 with the stock arm on the other build, as first reported.

RQ-N11, what the annealed placement says. H11 is inconclusive under the protocol text (no design excluded; the sign rule needs 12 of the 15 designs, so a relation defined for fewer than 12 designs cannot be supported): same-word pairs: median stratified distance difference -0.176 (one-sided Wilcoxon $p = 0.017$, n=12 designs; negative in at least two of three seeds on 9 designs, 12 needed; status inconclusive); same-module pairs: median stratified distance difference -0.188 (one-sided Wilcoxon $p < 0.001$, n=11 designs; negative in at least two of three seeds on 10 designs, 12 needed; status inconclusive). The variants of Table XII, labelled: with the sign rule scaled to $\lceil 12n/15 \rceil$ designs, which the first analysis code used and the protocol does not register (the previous primary status): word inconclusive (n=12), module supported (n=11), overall inconclusive; with the 95% exclusion of the amendment

TABLE IX

Round three: hypotheses, status and deciding number (all designs run; statuses use the rules in docs/plan/round3.md). H9 numbers are for the headline β_m of each source, Holm-adjusted over the three sources; a status that differs between all 15 designs and the 11 without the pilot designs is shown as pilot-sensitive.

RQ	Hypothesis	Status	Deciding number
N8	H8 packing dilutes structure	supported	median word purity 0.63, split 4.00 (n=15)
N10	H10a io-words add coverage	supported	median gain 24.1%, 15/15 designs
N9	H9a module purity up, CLBs within +5%	inconclusive (pilot-sensitive)	purity +0.000 (without pilot +0.035), CLB -1.6%, $p < 0.001$
N9	H9a word purity up, CLBs within +5%	supported	purity +0.111 (without pilot +0.120), CLB 0.0%, $p < 0.001$
N9	H9a word+io purity up, CLBs within +5%	supported	purity +0.104 (without pilot +0.124), CLB 0.0%, $p < 0.001$
N9	H9b module changes WL (analytic)	inconclusive (pilot-sensitive)	geomean 1.051, 3/15 better, $p = 0.090$; without pilot 1.069, $p = 0.006$
N9	H9b word changes WL (analytic)	inconclusive (pilot-sensitive)	geomean 1.032, 4/15 better, $p = 0.188$; without pilot 1.053, $p = 0.016$
N9	H9b word+io changes WL (analytic)	inconclusive (pilot-sensitive)	geomean 1.030, 4/15 better, $p = 0.188$; without pilot 1.050, $p = 0.016$
N9	H9c module+seed beats stock+seed	not supported	geomean 1.064, 2/15 better, $p = 1.000$
N9	H9c word+seed beats stock+seed	not supported	geomean 1.039, 4/15 better, $p = 1.000$
N9	H9c word+io+seed beats stock+seed	not supported	geomean 1.039, 5/15 better, $p = 1.000$
N10	H10b word+io beats word	not supported	geomean 0.998, 4/15 better, $p = 1.000$
N11	H11 same-word pairs closer than nets predict	inconclusive	median -0.176, $p = 0.017$, n=12; 9 of 12 designs sign-consistent, 12 needed; with the unregistered scaling (10 needed): inconclusive
N11	H11 same-module pairs closer than nets predict	inconclusive	median -0.188, $p < 0.001$, n=11; 10 of 11 designs sign-consistent, 12 needed; with the unregistered scaling (9 needed): supported
N11	H11 overall (both relations)	inconclusive	supported only if both are; unregistered scaling: inconclusive

TABLE X

RQ-N8 per design on the stock packing: logic CLBs, word purity (plurality share), the random-grouping null purity, word split factor (1.0 best), module purity, and the share of logic atoms lost to the front end (no word) and to packing (word atom not in its word's CLB).

Design	CLBs	Purity	Null	Split	Module	Lost: FE	Lost: pack
ch_intrinsics	13	0.86	0.51	3.00	1.00	54.7%	6.1%
stereovision3	15	0.48	0.25	2.00	1.00	40.4%	31.3%
diffeq2	22	0.96	0.31	2.00	1.00	31.8%	7.0%
diffeq1	37	0.80	0.27	7.50	1.00	37.4%	14.2%
mkPktMerge	21	0.86	0.75	1.00	0.92	7.3%	13.2%
raygentop	143	0.52	0.14	3.00	0.68	34.2%	32.6%
sha	142	0.47	0.19	11.50	1.00	50.5%	27.2%
boundtop	54	0.48	0.18	4.00	0.68	33.7%	34.2%
mkSMAdapter4B	125	0.63	0.20	5.00	1.00	51.3%	20.0%
or1200	252	0.64	0.20	5.50	0.76	51.1%	20.7%
stereovision0	707	0.27	0.06	4.00	0.74	22.1%	56.8%
stereovision1	788	0.36	0.08	4.00	0.71	17.1%	53.0%
mkDelayWorker32B	74	0.77	0.47	1.00	0.99	80.7%	4.3%
stereovision2	1956	0.59	0.09	6.00	0.88	4.9%	39.0%
blob_merge	494	0.91	0.09	1.00	1.00	30.5%	8.0%

d805789: word inconclusive (n=12), module inconclusive (n=11), overall inconclusive (scaled count: word inconclusive (n=12), module supported (n=11), overall inconclusive). Without the four pilot designs the protocol-text statuses are word inconclusive (n=9), module inconclusive (n=9), overall inconclusive; the literal 12 is unreachable there. Neighbour stability across VPR seeds (mean Jaccard of each CLB's four nearest neighbours): 0.267 against 0.058 for a permuted placement. The per-design "why are these neighbours" tables (direct nets, shared high-fanout net, carry chain) are in data/analysis/n11-why/.

V. Discussion

Three observations. First, the seed families span a narrow band but are not interchangeable: provenance (Urd) is the worst full seed in the soft arm (1.029) and the best in the hard arm, so it is not consistently better or worse than recovered clusters. The full and inter-cluster-only arms differ by 0.6% in pooled wirelength (1.009 vs 1.015); round one has no arm that randomizes the coarse cluster layout; the round-two permuted-cluster and random-start arms (Section IV-A) measure it. Second, VPR's checkpointing does not separate the crossover in

TABLE XI
RQ-N10: I/O words per design (buses of at least two bits) and the LUT word coverage without and with io-words.

Design	io-words	Word cov.	+ io-words
ch_intrinsics	8	16.4%	49.2%
stereovision3	3	26.0%	41.6%
diffeq2	8	31.8%	81.6%
diffeq1	11	21.5%	53.9%
mkPktMerge	5	49.7%	95.8%
raygentop	23	24.4%	57.8%
sha	4	2.5%	47.5%
boundtop	22	42.1%	64.3%
mkSMAdapter4B	29	12.6%	38.0%
or1200	36	27.6%	51.7%
stereovision0	27	32.1%	40.5%
stereovision1	20	50.3%	57.3%
mkDelayWorker32B	27	77.0%	78.1%
stereovision2	24	78.8%	85.2%
blob_merge	4	48.8%	52.2%

the measured range: the two sweeps find the same scale on 14 of the 15 designs where both found one, and the control extends it on stereovision0 (control 32 vs 16). RLPlace's prediction that checkpointing extends the useful range is therefore not observed on these designs; if it holds, its effect is smaller than the between-design variation. Third, the safe recipe is to cold-start from a cluster-derived placement at a temperature scale at or below the crossover, which is a multiple of VPR's own estimate; above it the seed's retention is still high per family (medians 0.52-0.74 at twice the crossover, individual runs 0.21-1.00) but it wins only 57 of 105 head-to-heads against an unseeded run at the same temperature – the coarse structure survives the anneal, the advantage does not.

Round two. The bitwise idea is tested in four forms, and the evidence is this: word clusters against module clusters are inconclusive (0.986); laying the words out bit-sliced against the quotient layout of the same clusters is inconclusive (0.982 in wirelength); the hierarchical combination is not supported as the best seed (rank 8 of 11); and seeding only the word cells is not distinguishable from seeding everything (H7d inconclusive, 0.980, better on 9 of 15). The word membership available from Odin III covers the flip-flops, adders, multipliers and memories but only 49% of the LUTs, because ABC's cone provenance merges the spans of mixed logic, so the word seeds are datapath-register seeds more than whole-datapath seeds; a result against bitwise seeding here is a result about that coverage as much as about the idea.

Every soft arm of Table IV, bitwise or not, random start included, lands between 0.980 and 1.029 of the baseline's wirelength: a band of 4.9% in a geomean over 15 designs. The conclusions below are conclusions about differences inside that band.

One confound the scale-1 comparison cannot remove: VPR derives its start temperature T_0 from the starting placement, so a start that is worse than the analytic one is annealed hotter and, in these runs, longer. The geomean start temperature relative to the baseline is 9.24 for the random start, 3.23 for Urd and 4.63 for the best new family, and placement runtime moves with it (Table IV). The arms are therefore compared at VPR's own temperature, not at equal anneal effort. The confound bears directly on four comparisons: H4 (random start, permuted and quotient layouts have T_0 of 9.24, 3.09 and 3.23 times the baseline's, for Urd), H7a (words 6.24 against Urd 3.23), H7b (slices 8.07 against words 6.24) and H7d (words_partial 4.63 against words 6.24; against the permuted Urd arm 3.09). The start temperatures of the arms differ by 3.1 to 9.2 times the baseline's, and the direction of the bias is unknown: a hotter start can erase more of the seed, or give the anneal more moves to repair a worse one. A random start beating a connectivity-aware start would be a statement about the schedule as much as about the start. The effort curve (N3) is the experiment that separates the two, and it stays open.

On the coarse-versus-fine question that round one left open, the arms give these numbers: H4 is not supported, with Urd at random 0.997, permuted 1.020 and quotient 1.029. RQ-N1 measures part of the round-one parity: VPR's final placement has 2.78× the null's module compactness and 3.79× its word compactness (medians), which is inconclusive as a rediscovery of the hierarchy by the NMI-and-compactness rule, which suggests the placer already groups much of what a module or word seed would group. Whether the structure of the netlist predicts where a seed helps is H2, inconclusive (the one correlation that reaches $p < 0.05$ is negative: the best new-family gain vs $z(Q)$; it tracks graph size, and with the edge count regressed out the partial ρ is -0.16, $p = 0.574$). All of this is $n = 15$ designs, median 125 CLBs, one VPR seed per configuration; the seed-to-seed spread of VPR is not measured here, so differences of about a percent should not be read as effects.

A. Round three discussion

Packing does dilute word structure (H8 supported): a word's atoms are spread over about 4.00 times the CLBs they would need, and the purity of the stock CLBs (0.63) is well above a random grouping (0.20) but far from word-pure. Of the logic atoms, 34.2% have no word at all (the front end) and a further 20.7% are word atoms packed away from their word (packing); these are two separate per-design medians, and the median front-end loss is larger than the median packing loss.

I/O words are a cheap way to recover part of the front-end loss (H10a supported: LUT word coverage 31.8% to 53.9%).

The Norns packer does what it was built to do: with the word affinity the word purity rises by 0.111 at a median CLB change of 0.0% (H9a supported; module inconclusive (pilot-sensitive), word+io supported). The module source has almost nothing to act on, because the designs are close to one module each (median module purity 0.99).

TABLE XII

RQ-N11, H11 under the protocol text and the labelled variants. n is the number of designs in which the relation is defined (and not excluded); median is the median over those designs of the stratified distance difference; k/need is the number of designs negative in two of three seeds over the number the rule needs; perm p is the permutation p of the median over designs (1000 label permutations pooled over designs for the protocol text and the scaled count, 200 for the amendment), reported and not used in the status. The status uses the median, the one-sided Wilcoxon test and k/need. The count of the pilot-free rows is shown scaled in the scaled rows only; with 11 designs the literal 12 cannot be met.

	Same word					Same module				
Variant	n	median	k/need	perm p	status	n	median	k/need	perm p	status
All 15 designs										
Protocol text, 12 of 15 designs (primary)	12	-0.176	9/12	<0.001	inconclusive	11	-0.188	10/12	<0.001	inconclusive
Scaled count (not registered)	12	-0.176	9/10	<0.001	inconclusive	11	-0.188	10/9	<0.001	supported
Amendment d805789, 12 of 15	12	-0.176	9/12	0.005	inconclusive	11	-0.188	10/12	0.005	inconclusive
Amendment d805789, scaled count	12	-0.176	9/10	0.005	inconclusive	11	-0.188	10/9	0.005	supported
Without the four pilot designs										
Protocol text, 12 of 15 designs (primary)	9	-0.127	6/12	<0.001	inconclusive	9	-0.141	8/12	<0.001	inconclusive
Scaled count (not registered)	9	-0.127	6/8	<0.001	inconclusive	9	-0.141	8/8	<0.001	supported
Amendment d805789, 12 of 15	9	-0.127	6/12	0.005	inconclusive	9	-0.141	8/12	0.005	inconclusive
Amendment d805789, scaled count	9	-0.127	6/8	0.005	inconclusive	9	-0.141	8/8	0.005	supported

TABLE XIII

Packer diagnostics per design at the headline β (seed 1). Matched: share of affinity-file entries that named an atom of the netlist (from the VPR log; the lowest per source: module 29.9% on mkDelayWorker32B; word 2.3% on mkDelayWorker32B; word+io 2.3% on mkDelayWorker32B). Queued: share of molecules handed to the candidate queue by the proposal step that entered it. 0-queued: share of proposal calls (one per cluster) that queued nothing. Columns in each group: module, word, word+io.

	Matched			Queued			0-queued		
Design	Mod.	Word	W+io	Mod.	Word	W+io	Mod.	Word	W+io
ch_intrinsics	100.0%	71.7%	63.8%	81.7%	86.9%	86.9%	1.0%	19.8%	19.8%
stereovision3	100.0%	100.0%	98.9%	66.3%	11.8%	11.8%	5.1%	84.2%	84.2%
diffeq2	86.1%	100.0%	100.0%	63.4%	74.7%	74.7%	6.1%	19.6%	22.0%
diffeq1	92.4%	100.0%	100.0%	53.2%	59.1%	59.1%	7.1%	27.8%	27.8%
mkPktMerge	100.0%	100.0%	54.1%	21.9%	23.8%	23.8%	58.2%	57.5%	57.5%
raygentop	95.1%	96.1%	96.4%	48.8%	42.9%	38.1%	13.1%	59.5%	59.8%
sha	94.4%	100.0%	100.0%	88.0%	45.6%	47.8%	4.1%	61.6%	57.0%
boundtop	90.2%	89.5%	90.1%	44.2%	63.0%	64.1%	15.9%	49.0%	47.6%
mkSMAdapter4B	96.5%	85.5%	88.7%	21.9%	49.6%	47.6%	20.5%	54.6%	47.4%
or1200	97.7%	97.6%	96.0%	29.1%	46.4%	42.1%	22.0%	52.4%	49.2%
stereovision0	100.0%	100.0%	99.8%	79.7%	77.6%	77.8%	6.8%	68.0%	67.8%
stereovision1	99.4%	99.3%	99.2%	34.1%	77.8%	76.0%	41.3%	44.4%	43.8%
mkDelayWorker32B	29.9%	2.3%	2.3%	7.9%	0.0%	0.0%	60.4%	100.0%	100.0%
stereovision2	73.0%	100.0%	99.9%	17.5%	46.1%	46.2%	57.4%	37.2%	37.0%
blob_merge	99.8%	100.0%	99.0%	92.4%	5.4%	12.9%	0.3%	98.5%	97.1%

Purer clusters did not give shorter wires. With the analytic start the word-affinity packing routes at 1.032 of the stock packer's wirelength (better on 4 of 15 designs, 90% interval 1.006 to 1.062), so H9b is inconclusive (pilot-sensitive): the sign is positive (worse) on every source, the two-sided test reaches the 5% level only without the four pilot designs, and the status is therefore not stated as a change. With the round-two word seed rebuilt on the new packing the ratio is 1.039 (H9c not supported): giving the round-two seed word-pure CLBs does not rescue it. The packing time grows by a factor 1.125 (word) and 1.318 (module). The seed noise (median coefficient of variation 2.4%, up to 9.7%) is of the same order as the effect, which is why the statistics use the design as the unit with three seeds each.

These runs do not remove the start-temperature confound of round two for the seeded arms, because a seeded start still carries its own start temperature; the analytic-start arm (H9b) does not have it, since the packing is the only difference.

In the median design both relations sit closer than their direct nets alone predict (stratified distance difference -0.176 for same-word pairs, -0.188 for same-module pairs), but not on every design: the same-word difference is positive on 3 designs (boundtop, raygentop, stereovision2) and the same-module difference on 1 designs (blob_merge). H11 is inconclusive under the protocol text, because its rule needs both relations to pass: same word, n=12, status inconclusive (negative in two of three seeds on 9 designs, 12 needed); same module, n=11, status inconclusive (10 designs, 12 needed). Two deviations of the analysis from the protocol text are reported beside it, not instead of it: the unregistered scaling of the count to $\lceil 12n/15 \rceil$ designs, which the first analysis used (r3stats.py), gives word

inconclusive (n=12), module supported (n=11), overall inconclusive; the amendment d805789, which excludes a design from a relation when 95% or more of its sampled pairs are related, for the word relation as well as the module relation, gives word inconclusive (n=12), module inconclusive (n=11), overall inconclusive. Both are recorded in the entry of 2026-10-25 of docs/plan/round3.md and in the Method. The neighbours are only partly stable: the mean Jaccard of each CLB's four nearest neighbours between VPR seeds is 0.267, against 0.058 for a permuted placement.

Round three leaves open the effort curve (N3), larger designs (N5), region looseness (N6), a seed-selector change in the Norns packer, and whether a different gain scale (the grid ended at the largest beta of the pilot list; the pilot did not discriminate the betas, so the grid is not informed by it) changes the sign of H9b. The packing time ratios of Stage 4 compare two builds and the proposal step has the counting and ordering limits described in the Method, both to be fixed in round 4.

VI. Conclusion

Seeding VPR's anneal is worth it only as a calibrated cold start: at VPR's own start temperature the cluster layouts and the provenance hierarchy place at parity with the analytic start, but from a good final placement they keep winning when the anneal is restarted hotter, up to the largest tested scale that still wins – a lower bound on the fall point where the sweep hit its cap. The recipe is that crossover table, not a seed-specific tuning: start at or below it, and expect the seed to inform the coarse placement while the anneal supplies the fine one.

Round two adds bitwise seeds to the provenance hierarchy. At VPR's own start temperature the best of the four new families, words_partial, is 0.994 of the baseline's wirelength; that is a figure selected as the best of four on the data it is reported on, it sits inside the 0.980 to 1.029 band that every soft arm falls in, and it is not an effect. The question that made the idea worth testing, bit-level structure against module structure, comes out inconclusive (H7a) and not supported (H7c) on these designs.

Appendix

This log records the thought pattern of the study so a reader can follow how each result produced the next question. Entries are dated; numbers are those of the generated tables.

a) 2026-10-23, round one: the question and its hypotheses.: Does seeding VPR's anneal from netlist clusters or the provenance hierarchy improve placement, and how hot can the anneal start before it destroys the seed? Hypotheses: a seed helps at VPR's own start temperature; the helpful temperature range is bounded by a design-level crossover; the provenance hierarchy and recovered clusters differ; hard regions help or hurt.

b) 2026-10-24, what round one said.: Soft seeds sat at parity with the analytic baseline (wirelength 1.009 full, 1.015 inter-cluster only; channel width 0.973; runtime 1.087); hard regions were worse (1.088, 14 of 104 better); seed retention was 0.68/0.70 at scale 1 but a seeded run won only 57 of 105 head-to-heads at twice the crossover; Urd was the worst soft and the best hard seed.

c) 2026-10-24, what it suggested.: The baseline is VPR's connectivity-aware analytic start, so better than random was never tested; the anneal budget may erase what a seed adds (an effort curve is untested, N3); VPR's flat placement may already rediscover the structure the seed carries (N1); the designs are small, median 125 CLBs (size effect untested, N5); and whether the gain is the clustering or the coarse layout was not separable (N4).

d) 2026-10-24, Peter's bitwise idea.: "consider that bitwise is more relevant in the Norn path so we should consider that as an additional seeding aspect (not necessarily the entire HDL structure (though hierarchically yes)). Let's include this and run the next set of experiments noting this thought pattern in the appendix of each paper so we can see the progress ... write up into the papers findings." Reading: Norns is the datapath path, bits are the unit of structure there, so word and bit-slice structure is a seeding aspect of its own, with the module hierarchy kept at the coarse level.

e) 2026-10-24, round two: questions and hypotheses.: N7 (bitwise seeds): H7a to H7d. N1: H1 (VPR rediscovers the hierarchy). N2: H2 (structure predicts the gain). N4: H4 (random start, permuted clusters, quotient layout, analytic start). The word membership had to come from data: the provenance BLIF line is a statement, so the membership is read from Odin III's bus names and source spans and joined by net id (Section III-G); only about 49% of the LUTs get a word.

f) 2026-10-24, round two: outcomes.: Inconclusive for H7a, inconclusive for H7b, not supported for H7c, inconclusive for H7d, not supported for H4, inconclusive for H1 and inconclusive for H2; the numbers are in Table XV and Section IV-A.

g) 2026-10-24, rules changed after the data.: Two decision rules were touched after the results were visible, and one count was corrected. (1) H2: the rule fixed in the methodology text was "not supported if no correlation reaches $p < 0.05$". When the full data showed one significant correlation, a negative one (best new-family gain against $z(Q)$, $\rho = -0.59$, $p = 0.021$ with the null of the first run), the rule was briefly changed to count only positive correlations, which turned the status into not supported. That change is reverted: under the rule as first written the status is inconclusive. The null was then recomputed without the cap on edge swaps (RQ-N2) and the correlation is $\rho = -0.58$, $p = 0.023$. (2) H7d: the first rule called "at least as good" supported at a geomean ratio of 1 or below, with no test and no margin, and so called H7d supported at 0.980 (better on 9 of 15, $p = 0.607$).

TABLE XIV
How much of each design the words_partial seed places. Logic and memory CLBs only (pads and multipliers are anchored in every arm); Seeded = CLBs in a word cluster that VPR's initial placement is given a slot for; Moved = CLBs whose seeded position differs from VPR's analytic start, for words_partial and for the full words seed.

Design	Placed blocks	Logic+mem.	Seeded	Moved (partial)	Moved (words)
ch_intrinsics	243	14	12	11	11
stereovision3	56	15	11	11	12
diffeq2	189	22	16	15	20
diffeq1	300	37	33	33	35
mkPktMerge	503	36	35	25	35
raygentop	691	144	132	132	141
sha	216	142	130	130	142
boundtop	444	55	53	53	53
mkSMAdapter4B	528	130	96	94	124
or1200	1034	254	158	155	247
stereovision0	1073	707	696	696	707
stereovision1	1091	788	786	785	785
mkDelayWorker32B	1176	117	22	18	88
stereovision2	2479	1956	1952	1948	1956
blob_merge	630	494	321	321	492

TABLE XV
Research questions and hypotheses of the study with status and the deciding number.

RQ	Hypothesis	Round	Status	Deciding number
R1	soft seed stays at parity	1	supported	wirelength 1.009
R1	hard regions help	1	not supported	1.088 (14/104)
R1	seeds win at 2× crossover	1	not supported	57/105
N7	H7a words beat Urd	2	inconclusive	0.986 (10/15)
N7	H7b slices beat words	2	inconclusive	WL 0.982, ch. 1.033
N7	H7c hier is best	2	not supported	rank 8/11
N7	H7d partial better than full	2	inconclusive	0.980
N4	H4 random < permuted < quotient	2	not supported	0.997/1.020/1.029
N1	H1 baseline already coherent	2	inconclusive	8/11 mod., 9/15 word
N2	H2 structure predicts gain	2	inconclusive	0+/1- of 9 at $p < 0.05$

A geomean near 1 with a non-significant sign test is no evidence either way, so the status is inconclusive ("no evidence of a difference"). (3) H1: the count of designs above the null in compactness was first divided by the number of designs scored on NMI; it now has its own denominator, and the module row no longer counts the designs with fewer than two real modules (13 designs became 11). The status of H1 did not change.

h) Still open.: N3 (effort curve: how the result changes with the anneal budget, e.g. inner_num); N5 (size: designs of 1000 CLBs and up, where a seed has more to say); N6 (region looseness: constraint regions between the soft seed and the tight regions of round one); LUT coverage of the word membership (a front-end change, not a seeding change); more than one VPR seed per configuration. Next round: H7d is to be tested as non-inferiority, which needs a stated margin on the geomean wirelength ratio; the margin is to be fixed, together with every other decision rule, in a commit before any run, and chosen from the run-to-run spread that several VPR seeds per configuration will measure. No margin is proposed from the present single-seed data.

i) 2026-10-24, round three: Peter's two questions.: "For Norns, one question is does the clustering step of FPGA mapping become the problem - how is the initial seeding for an hdl construct get clustered? We might have to make our version of clustering for this information (how is it done - just with TVpack?). Second, if we map designs and have provenance (including the clustering) what do we learn from the placed (just by aneealing results) can we make some questions like (what is close to one another and using the provenance info (why are things close to one another?)" and "Also, the outputs (words) and the inputs as words also provide hints of where the words are in norns. Let's build our own Norns based packer from scratch (include it in Norns as a copy/modification to Apack to allow us to our controlled experiements)." Reading: the clustering step (VPR's AAPack, not T-VPack) may be where the HDL structure is lost before placement starts; the I/O buses are words too, and say where words sit; and the annealed placement, read with provenance, may answer which constructs end up near each other and why.

j) 2026-10-24, what round two suggested.: Every soft seed stayed within 0.980 to 1.029 of the baseline: the seeds were applied after packing, to whole CLBs, and ABC had already merged cones (only about half of the LUTs get a word), so packing dilution was untested (N8) and word structure at the pad level was unused (N10). The T0 confound of round two (a seeded start also changes VPR's start temperature) is still open for the seeded arms: round three does not remove it. The analytic-start comparison (H9b) addresses it for the packing question, because there the packing is the only difference.

k) 2026-10-24, round three: questions and hypotheses.: N8: H8. N10: H10a, H10b. N9: H9a, H9b, H9c with the Norns packer. N11: H11. All rules, margins and the grid-selection rule are fixed in docs/plan/round3.md, committed as 4fb8b60 before any measurement.

l) 2026-10-24, round three: outcomes.: H8 supported; H10a supported; H9a module inconclusive (pilot-sensitive), word supported, word+io supported; H9b module inconclusive (pilot-sensitive), word inconclusive (pilot-sensitive), word+io inconclusive (pilot-sensitive); H9c module not supported, word not supported, word+io not supported; H10b not supported; H11 inconclusive (word inconclusive, module inconclusive). The numbers are in Table IX.

m) 2026-10-25, round three fix round (after review).: The rule-provenance description was corrected: the entry

d805789, called a clarification, is an amendment that changed two H11 rules (label permutations; the 95% exclusion), and two designs had Stage-4 runs finished before it was committed (Method). H11 is now given under the protocol text (no exclusion, 12 of the 15 designs, 1000 pooled permutations) as the primary status: word inconclusive, module inconclusive, overall inconclusive; the status of the first analysis (the sign rule scaled to $\lceil 12n/15 \rceil$ designs, not registered) was word inconclusive, module supported, overall inconclusive, and the amendment's exclusion gives word inconclusive, module inconclusive. The n=11 pass was added for H8, H10a and H11, Holm now keeps $m = 3$ (no status changed), and the positional null uses 200 permutations. The packer got a log-only proposal counter and an integer-parse error; the packing is unchanged (Method), and the per-design matched-entry and proposal tables are in Table XIII. Not done: the proposal-counting behaviour (round 4).

n) 2026-10-24, next.: The effort curve (N3), larger designs (N5), a seed-selector change and a larger gain scale in the Norns packer, and the packing-time cost of the affinity term. Added 2026-10-25: fix the proposal counter (count queued molecules only; return a bool from the candidate-queue step) and the scan order of the group members, then rerun the packing experiments.

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TABLE XVI
Round-three research questions and hypotheses with status and the deciding number (extends Table XV).

RQ	Hypothesis	Round	Status	Deciding number
N8	H8 packing dilutes structure	3	supported	median word purity 0.63, split 4.00 (n=15)
N10	H10a io-words add coverage	3	supported	median gain 24.1%, 15/15 designs
N9	H9a module purity up, CLBs within +5%	3	inconclusive (pilot-sensitive)	purity +0.000 (without pilot +0.035), CLB -1.6%, $p < 0.001$
N9	H9a word purity up, CLBs within +5%	3	supported	purity +0.111 (without pilot +0.120), CLB 0.0%, $p < 0.001$
N9	H9a word+io purity up, CLBs within +5%	3	supported	purity +0.104 (without pilot +0.124), CLB 0.0%, $p < 0.001$
N9	H9b module changes WL (analytic)	3	inconclusive (pilot-sensitive)	geomean 1.051, 3/15 better, $p = 0.090$; without pilot 1.069, $p = 0.006$
N9	H9b word changes WL (analytic)	3	inconclusive (pilot-sensitive)	geomean 1.032, 4/15 better, $p = 0.188$; without pilot 1.053, $p = 0.016$
N9	H9b word+io changes WL (analytic)	3	inconclusive (pilot-sensitive)	geomean 1.030, 4/15 better, $p = 0.188$; without pilot 1.050, $p = 0.016$
N9	H9c module+seed beats stock+seed	3	not supported	geomean 1.064, 2/15 better, $p = 1.000$
N9	H9c word+seed beats stock+seed	3	not supported	geomean 1.039, 4/15 better, $p = 1.000$
N9	H9c word+io+seed beats stock+seed	3	not supported	geomean 1.039, 5/15 better, $p = 1.000$
N10	H10b word+io beats word	3	not supported	geomean 0.998, 4/15 better, $p = 1.000$
N11	H11 same-word pairs closer than nets predict	3	inconclusive	median -0.176, $p = 0.017$, n=12; 9 of 12 designs sign-consistent, 12 needed; with the unregistered scaling (10 needed): inconclusive
N11	H11 same-module pairs closer than nets predict	3	inconclusive	median -0.188, $p < 0.001$, n=11; 10 of 11 designs sign-consistent, 12 needed; with the unregistered scaling (9 needed): supported
N11	H11 overall (both relations)	3	inconclusive	supported only if both are; unregistered scaling: inconclusive